E33: solver B's pivot selection, counted (and two cheaper selections)

Run report and results, 2026-11-01. Research note `research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md`: §9.35 (pre-registration, commit `27663cc4d7`, before any code) and §9.37 (results). Code `c443138920` (unit-test compile fix and checks example `c689d19821`; library build unchanged). Binaries pinned by sha256 in `experiments/37_solver_b_pivot_count/binary_sha256.txt`. Records: `experiments/37_solver_b_pivot_count/` (`stage_a.{json,log}` with the JSON in the artifact store, sha256 `7bc6e00b…`; `e2e_*_karatsuba_p*s*.json` and logs; `identity_existing_solvers.json`; `checks.log`; `tables.md`; `summary.json`). Host: the 4-vCPU Claude Code container, at most two cores. isolab-eks is not reachable from it (no AWS CLI; the GitHub route was refused through the proxy). No isolation tier is earned: operation counts only, no wall-time claim.

What ran

- **Stage A:** E31′s calls exactly. On each of E25′s ten GLS curves ($p \in \{101, 149, 197, 269, 397\}$, seeds 1, 2), the first 1,000 targets of E25 arm A's stream, one thread. Seven solvers ran per call from the same per-target seed: the existing solver (E25), A, A counted, B, B counted, B′ and B″. That is 70,000 solves, 0 failures.
- **End to end:** E27′s command (E31′s pipeline with Karatsuba $\mathbb{F}_{p^4}$ products) with B counted, B′ and B″, on the ten curves, both arms, `RAYON_NUM_THREADS=4`. 30 runs, 0 failures. B″ was carried by the registered rule: its Stage A $S(0.1)$ was below A's on every curve.
- **Identity:** the pre-E33 binary and E33′s at $p = 101$ seed 2 for `macaulay`, `chain`, `chain-sparsest` and `chain --karatsuba`. The checks example `e33_checks` (32 calls, two negative controls).

Counting rule (registered)

One native operation per Macaulay entry examined for its zero status and one per key comparison, priced at $g \cdot 97$ $\mathbb{F}_p$ multiplications, $g \in \{0, 0.1, 1\}$. B: an initial count of every entry, $|\text{bucket}| - 1$ comparisons per contested column, and a recount of cols − lead entries per updated row. B′: counts inherited from the equations, plus two status tests per entry the pivot row touches and one per row. B″: inherited counts and the Markowitz fill bound, at two operations per updated row. Lead scans are common to every rule and are emitted separately (sensitivity only).

Measured

- **Identity (G2.124):** all 10,000 Stage A calls agree with the existing solver for every solver. Phase counts are equal (A counted = A, B counted = B′ = B), the B′ digests equal B's, and B″'s maintained counts never differ from a recount. All 30 end-to-end records keep every non-product field of E31 B's records (`folded.recovered` is a key added after E31). The old and new binaries give identical records for the four existing configurations. Both negative controls were detected on 32/32 calls. The in-crate unit tests type-check but were not executed (memory; see Limits).
- **Selection work per call** (Stage A; end to end agrees): B 1.049–$1.057 \cdot 10^6$ (initial 176,064, comparisons 3,680–3,713, recounts 8.69–$8.77 \cdot 10^5$); B′ 2.93–$2.98 \cdot 10^5$ (0.279–0.282 of B); B″ 12,696–12,803. B″'s C median is 1.205–1.221 × B's.
- **End to end** (20 rows), as ranges of S_{E25}/S_X, S_A/S_X and S_X/rho at 0 / 0.1 / 1 GAE:
  - B: E25 4.53–4.76 / 0.274–0.276 / 0.157–0.158; A 1.82–1.86 / 0.193–0.197 / 0.149–0.153.
  - B′: E25 4.53–4.76 / 0.679–0.683 / 0.404–0.407; A 1.82–1.86 / 0.478–0.487 / 0.384–0.393.
  - B″: E25 3.83–3.98 / 1.457–1.497 / 0.981–0.997; A 1.538–1.558 / 1.046–1.048 / 0.9465–0.9480.
  - Distance to the matched rho (walk (iii) of E27): at 0 GAE, B and B′ $3.45 \cdot 10^3$–$2.19 \cdot 10^5$ and B″ $4.12 \cdot 10^3$–$2.61 \cdot 10^5$; at 1 GAE, B″ $9.58 \cdot 10^4$–$7.01 \cdot 10^6$. Rho is ahead of every row at every weight.

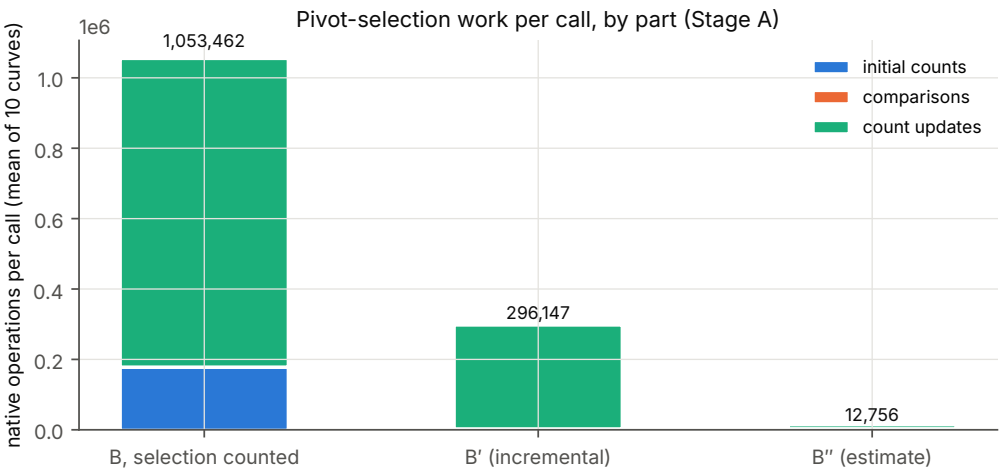


Figure 1: Pivot-selection native operations per call by part, mean of the ten curves' Stage A means (1,000 calls each). B″'s comparisons (≈ 4,100) and updates (≈ 8,150), and B′'s initial counts (≈ 506), are too small to see at this scale; exact values are in `tables.md`. Source: `summary.json`.

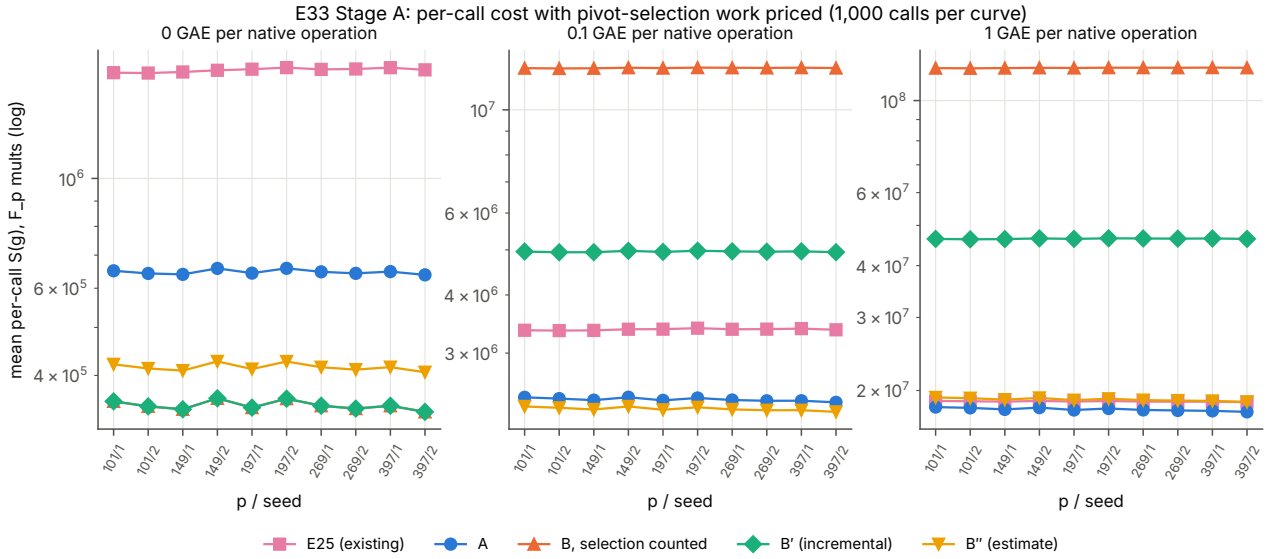


Figure 2: Stage A mean per-call $S(g) = C + 97g(\text{entries} + \text{selection})$ per curve, for E25, A, B (selection counted), B′ and B″, at 0, 0.1 and 1 GAE. At 0 GAE, B and B′ coincide (same pivots, same C). Source: `summary.json` (from `stage_a.json`).

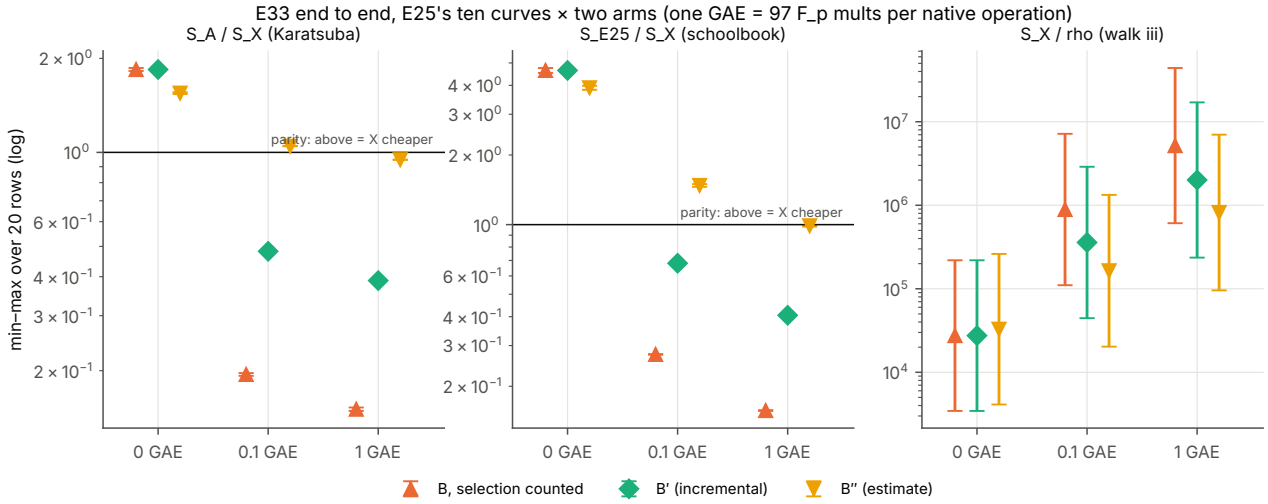


Figure 3: End to end, min–max over the 20 rows (marker at the geometric midpoint). S_A/S_X uses E27′s solver-A records (Karatsuba); S_{E25}/S_X uses E25′s records in schoolbook arithmetic; S_X/rho uses walk (iii). Above parity, X is cheaper. Source: `summary.json`.

Verdicts (pre-registered, §9.35)

G2.124 holds (the unit tests were not executed; disclosed). G2.125, G2.126, G2.127, G2.128, G2.129, G2.130, G2.132 and G2.133 hold. G2.131 is **falsified** against E25 at 1 GAE: B″ is behind E25 on all 20 rows (0.981–0.997). It holds against A.

Ranks (only where the order holds at 0, 0.1 and 1 GAE on every row)

- B, B′, B″ against A: **unresolved**. Each is ahead at 0 GAE. B and B′ are behind from 0.1 GAE; B″ is ahead at 0.1 GAE and behind at 1 GAE. A remains the ranked solver.
- B, B′, B″ against E25: **unresolved** (each is behind E25 at 1 GAE).
- Rho is ahead of B, B′ and B″ across the band.

What it says, and what it does not

Counting B's selection is accounting, not an improvement. It settles what §9.24 left open: B's advantage exists at 0 GAE only. Its pass reads the matrix about six times per call, which is far above the validator's lower bound of one read. B′ keeps B's choices at 28 % of the work and is still behind. B″ is cheaper than A with its own selection work counted at 0 and 0.1 GAE. At 1 GAE its $1.27 \cdot 10^4$ selection operations ($1.2 \cdot 10^6$ multiplications) exceed its $2.3 \cdot 10^5$ saving. It is therefore not ranked, and not claimed as an improvement across the band. The band at 1 GAE is set by the 176,064 Macaulay entries per call, which no pivot rule removes.

Limits

- The four in-crate unit tests (E31′s and E33′s in `klein_v4::tests`) type-check (`cargo check --release --lib --tests`) but were not run. A release test build of the library needs about 9 GB of RSS. The first attempt pushed the shared container's memory cgroup over its limit, and the kernel killed it together with a parallel experiment's process (E32′s `e32_ladder`). The solver-level test runs instead as `examples/e33_checks.rs`. The echelon-level test is covered only indirectly, by equal phase counts and outputs on 10,000 calls.
- E25′s lead scans in the sensitivity are A's (same echelon, same rows). The existing solver's other unpriced traffic (normal forms, Hessenberg) is not emitted for any arm.
- Not validated: no replay certificate or validator report yet, so no leaderboard or `benchmarks/` row (validation first, per the task).